

EXP.NO: 7	Producer Consumer Problem Solution using Semaphore
DATE: 06/03/2026	

AIM:

To implement the Producer Consumer problem using semaphores in Linux.

ALGORITHM:

1. Initialize semaphores `mutex`, `full`, and `empty`.
2. Producer produces items and inserts them into buffer.
3. Consumer consumes items from buffer.
4. `wait()` and `signal()` operations control synchronization.
5. Repeat until all items are produced and consumed.

CODE:

```
#include <stdio.h>
#include <pthread.h>
#include <semaphore.h>
#include <unistd.h>

sem_t empty, full, mutex;
int x = 0;

void *producer(void *p)
{
    for(int i=1; i<=5; i++)
    {
        sem_wait(&empty);
        sem_wait(&mutex);

        x++;
        printf("Producer produces item %d\n", x);

        sem_post(&mutex);
        sem_post(&full);

        sleep(1);
    }
}

void *consumer(void *p)
{
    for(int i=1; i<=5; i++)
    {
        sem_wait(&full);
        sem_wait(&mutex);
```

```

        printf("Consumer consumes item %d\n", x);
        x--;

        sem_post(&mutex);
        sem_post(&empty);

        sleep(1);
    }
}

int main()
{
    pthread_t p, c;

    sem_init(&empty,0,5);
    sem_init(&full,0,0);
    sem_init(&mutex,0,1);

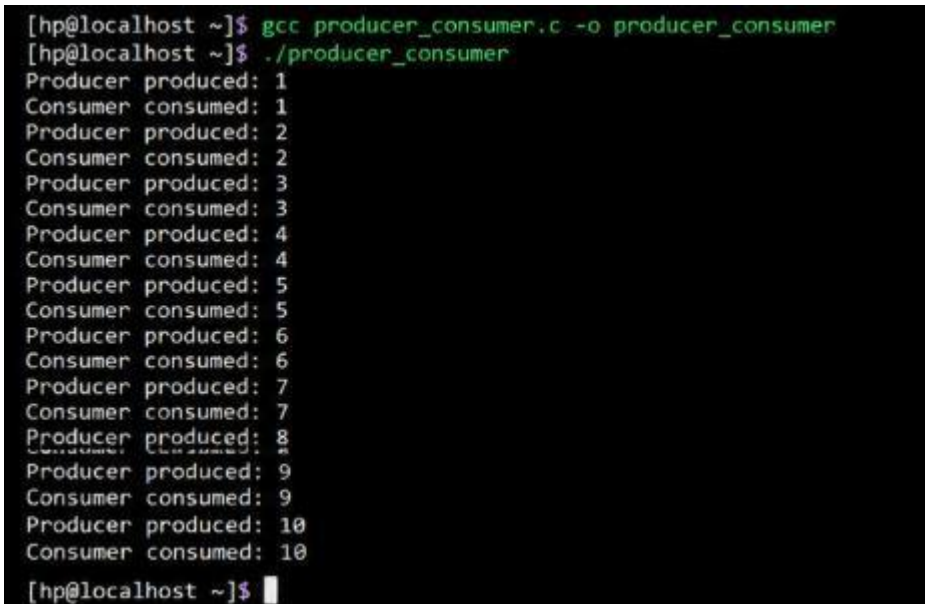
    pthread_create(&p,NULL,producer,NULL);
    pthread_create(&c,NULL,consumer,NULL);

    pthread_join(p,NULL);
    pthread_join(c,NULL);

    return 0;
}

```

OUTPUT:



```

[hp@localhost ~]$ gcc producer_consumer.c -o producer_consumer
[hp@localhost ~]$ ./producer_consumer
Producer produced: 1
Consumer consumed: 1
Producer produced: 2
Consumer consumed: 2
Producer produced: 3
Consumer consumed: 3
Producer produced: 4
Consumer consumed: 4
Producer produced: 5
Consumer consumed: 5
Producer produced: 6
Consumer consumed: 6
Producer produced: 7
Consumer consumed: 7
Producer produced: 8
Producer produced: 9
Consumer consumed: 9
Producer produced: 10
Consumer consumed: 10
[hp@localhost ~]$

```

RESULT:

Thus, the Producer Consumer problem using semaphores was implemented and executed successfully.